CI-Net: Clinical-Inspired Network for Automated Skin Lesion Recognition

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***Abstract*—The lesion recognition of dermoscopy images is significant for automated skin cancer diagnosis. Most of the existing methods ignore the medical perspective, which is crucial since this task requires a large amount of medical knowledge. A few methods are designed according to medical knowledge, but they ignore to be fully in line with doctors' entire learning and diagnosis process, since certain strategies and steps of those are conducted in practice for doctors. Thus, we put forward Clinical-Inspired Network (CI-Net) to involve the learning strategy and diagnosis process of doctors, as for a better analysis. The diagnostic process contains three main steps: the zoom step, the observe step and the compare step. To simulate these, we introduce three corresponding modules: a lesion area attention module, a feature extraction module and a lesion feature attention module. To simulate the distinguish strategy, which is commonly used by doctors, we introduce a distinguish module. We evaluate our proposed CI-Net on six challenging datasets, including ISIC 2016, ISIC 2017, ISIC 2018, ISIC 2019, ISIC 2020 and PH2 datasets, and the results indicate that CI-Net outperforms existing work. The code is publicly available at https://github.com/lzh19961031/Dermoscopy_classification.**

***Index Terms*—Skin lesion recognition, clinical-inspired, dermoscopy images, deep learning, neural network.**

I. Introduction

SKIN cancer is a common disease, with increasing death rates in recent years, and has aroused worldwide attention [1], [2], [3]. Shown in Fig 1, countries exposed to higher levels of ultraviolet radiation, which are mostly located tropically, experience significantly higher rates of skin cancers than others [4], [5]. Australia had the highest overall rate of skin cancer, followed by New Zealand, indicating that skin cancer should be taken seriously. Dermoscopy [6], [7], as a recent non-trauma skin imaging technology, is widely used and an essential means of enhancing diagnostic performance.

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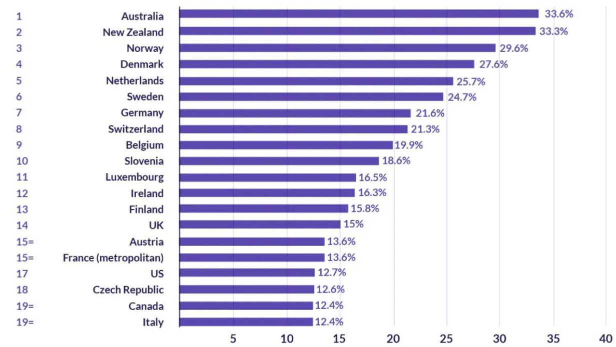

Fig. 1. The information of 20 countries with the highest skin cancer rates in the world.

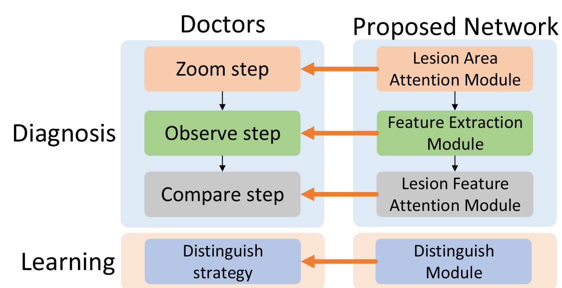

Fig. 2. The relationship between doctors' diagnostic, learning process and our network.

However, manual inspection by dermatologists requires high degrees of skill and is an exhaustive time-consuming task. Besides, it is subjective and bias-prone to human error. Thus, numerous algorithms have been dedicated to tackling automatic skin lesion recognition on dermoscopy images, so as this work.

Early methods extract hand-crafted features, including color, texture, shape specifically for skin lesion [8], [9], [10], [11], [12], and then conduct the classification. With the current trend of deep learning, CNN-based methods have established an overwhelming presence [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23]. Most of them do not intend to involve medical knowledge and therefore treat this task as same as the natural image recognition. For example, MRCN [24] argues to model the relationship both from the region level and the image level to better analyze the dermoscopy data. Besides, [13] utilizes data from the two modalities to obtain more comprehensive information and facilitate the final classification. A two-branch network is proposed in [25], which conducts skin lesion segmentation and classification simultaneously, to select a better feature. To ease the issue of insufficient data, a GAN-based model is introduced in [26] to synthesize new dermoscopy images, and serve as a kind of augmentation.

Despite the impressive breakthroughs brought by these approaches, they neglect the fact that this task requires medical expert knowledge to be accomplished, and inevitably ignore infusing this valuable knowledge into their method. This would lower the explainability of the method and limit the performance. To overcome this problem, there has been a growing interest in facilitating the CNN by involving doctor's knowledge [27], [28], [29], [30], [31]. For example, DermESy [31] incorporates ABCD rule for dermoscopy image classification, which is inspired by clinical knowledge. Representations useful for clinical diagnosis are proposed in [30], including structure, shape and color, to better leverage the medical criteria. Even though these methods integrate doctors' knowledge, they have not fully taken advantage of doctors' entire diagnosis process and learning strategy specifically, since this knowledge could be beneficial for deep incorporation between medical perspective and CNNs. This prevents them from showing better clinical interpretability and usability. Thus, involving clinical protocols and enabling CNNs to be more in line with doctors' expertise, i.e., simulating all the processes and strategies of professionals, becomes more and more appealing.

Our conference paper CIN [32] tackles this problem by utilizing two sources of information: **(1) doctors' diagnostic process** and **(2) doctors' learning strategy**. The diagnostic process of dermatologists consists of three main steps: **i. zoom step:** zoom in and focus on the central lesion area. **ii. observe step:** observe the lesion area and extract the feature. **iii. compare step:** compare the important local-attended regions within the lesion area and mine the correlation. To mimic these three steps, a lesion area attention module, a feature extraction module and a lesion feature attention module are proposed respectively. For the learning process of doctors, **distinguish strategy** is widely applied, which is to simultaneously learn from multiple images, and distinguish whether they are from the same class. To simulate this strategy, a distinguish module is introduced. Nonetheless, the incorporation of the above knowledge and CNN is not thorough enough for two reasons. Firstly, the lesion feature attention module in CIN, which aims to mimic the compare step, only conducts spatial-wise attention mechanisms, i.e., models the relationship between different feature elements. However, it ignores that channel-wise analysis could also facilitate the network since feature channels are rich in region information [33], [34]. Secondly, the distinguish module, which simulates the distinguish strategy, has a simple architecture, i.e., several conv layers, thus it may have limited representation learning ability.

In this paper, we put forward a Clinical-Inspired Network (CI-Net) to further leverage the information of doctors' diagnostic and learning process, and improve our previous work CIN [32]. The relationship between doctors' diagnostic, learning process and our network is illustrated in Fig. 2. Based on CIN, we improve the architecture of the lesion feature attention module and the distinguish module, while maintaining the lesion area attention module and the feature extraction module. The lesion feature attention module improves the analysis ability for the local-region information and involves the medical expertise more thoroughly. It extends from only spatial attention in [32] to both spatial-wise attention and channel-wise attention. The two attentions first squeeze the information channel-wisely and spatial-wisely, then highlight the useful information. The distinguish module improves the representation analysis and distinguishing ability. It extends from only several learnable layers in [32] to representation analysis from both channel-wise and spatial-wise. Experiments are conducted on six datasets, and the consistent state-of-the-art performance proves our method's effectiveness.

Overall, this work makes the following contributions: (1) A novel Clinical-Inspired Network (CI-Net) is introduced to mimic professional dermatologists' diagnostic process and learning strategy. Compared to other clinical inspired approaches, our work simulates these processes and strategies more comprehensively and completely. In addition, we utilize the clinical protocol of skin disease to explain the visualization result, in order to qualitatively show the clinical interpretability and usability of our method. (2) Two new modules are proposed, including a lesion feature attention module and a distinguish module, to infuse the expert knowledge into our neural network. (3) Our CI-Net achieves state-of-the-art performance both quantitatively and qualitatively on six benchmark datasets: ISIC 2016, ISIC 2017, ISIC 2018, ISIC 2019, ISIC 2020 and PH2.

This work extends our conference paper [32] with the following distinctions made. (1) A major change of the architecture of the lesion feature attention module and the distinguish module, by adding more comprehensive analysis from more perspectives. (2) More ablation study experiments are conducted on more datasets, including the investigation of basic configurations of the model, individual modules and the effect of the proposed modules. The results prove the effectiveness of the experimental settings and the architectural design. (3) A major expansion of benchmark study experiments, which expands from two benchmark datasets to six. Our CI-Net achieves state-of-the-art results on all of them, proving its robustness.

The rest of the paper is organized as follows. Section II describes the related work. Section III introduces the specific network architecture. Section IV describes the experiment settings and results. Section V presents our conclusions.

II. Related Work

In this section, we will first introduce hand-crafted, CNN-based methods and clinical-inspired methods for skin lesion classification. After that, we will review some related works of attention mechanisms.

A. Hand-Crafted Methods for Skin Lesion Classification

A large amount of low-level hand-crafted features have been proposed for skin lesion classifications, including texture [10], shape [11], color [9] and the combination after selecting proper features among them [12]. Some other attempts include

TABLE I
DETAILS OF EACH DATASET IN EXPERIMENTS

Dataset	Number of category	Number of images for each class	Evaluation Protocol
ISIC 2016 [45]	2	1029(Non-melanoma) & 248(Melanoma)	900 train, 379 test (Official split)
ISIC 2017 [46]	3	491(Melanoma) & 344(Seborrheic keratosis) & 1765 (Nevus)	2000 train, 600 test (Official split)
ISIC 2018 [47], [48]	7	1113(Melanoma) & 6705(Melanocytic nevus) & 514(Basal cell carcinoma) & 327(Actinic keratosis) & 1099(Benign keratosis) & 115(Dermatofibroma) & 142(Vascular lesion)	10015 train, 1512 test (Official split)
ISIC 2019 [46]–[48]	8	4522(Melanoma) & 12875(Melanocytic nevus) & 3323(Basal cell carcinoma) & 867(Actinic keratosis) & 2624(Benign keratosis) & 239(Dermatofibroma) & 253(Vascular lesion) & 628(Squamous cell carcinoma)	25531 train, 8238 test (Official split)
ISIC 2020 [49]	2	32542(Non-melanoma) & 584(Melanoma)	33126 train, 10982 test (Official split)
PH2 [50]	2	160(Non-melanoma) & 40(Melanoma)	2000 train(ISIC2017), 200 test(PH2) (Official split)

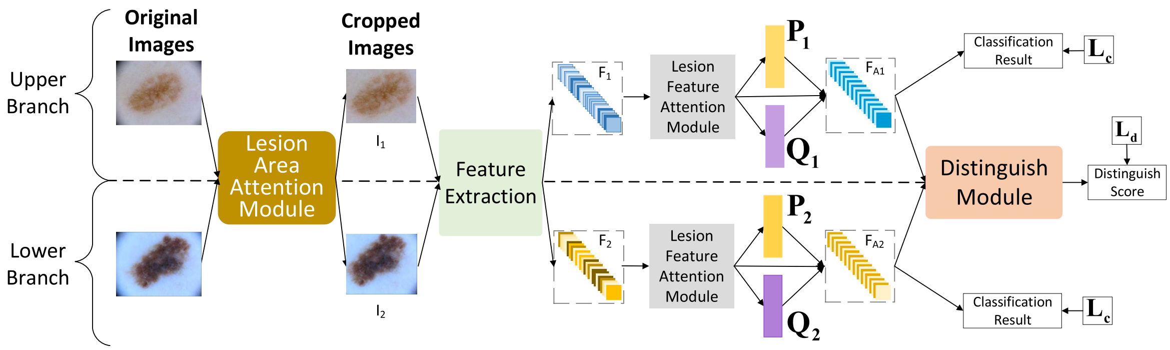

Fig. 3. An overview of the CI-Net. There are two branches, the upper branch and lower branch, and four modules: the lesion area attention module, the feature extraction module, the lesion feature attention module, and the distinguish module.

TABLE II
RESULTS ON ISIC 2016 DATASET. "AP" IS THE GOLD EVALUATION METRIC, HIGHLIGHTED BY "*" IN THE UPPER RIGHT CORNER

Method	AP^*	ACC	AUC
CI-Net	**0.815**	**0.904**	**0.904**
Teledermatology [52]	-	0.903	-
MRCN [24]	0.807	0.898	0.900
GP-CNN-DTEL [53]	0.728	0.863	0.861
L-CNN [23]	0.724	0.876	0.854
Taxonomies [29]	0.698	-	0.838
DermaKNet [28]	0.697	-	0.832
DCNN-FV [19]	0.685	0.868	0.852
SDL [15]	0.664	0.858	0.818
CUMED [17]	0.637	0.855	0.804
GTDL [45]	0.619	0.813	0.802
Result2-3B [45]	0.615	0.844	0.808
USYD [45]	0.580	0.686	0.793
Mufic-IT [45]	0.534	0.760	0.685

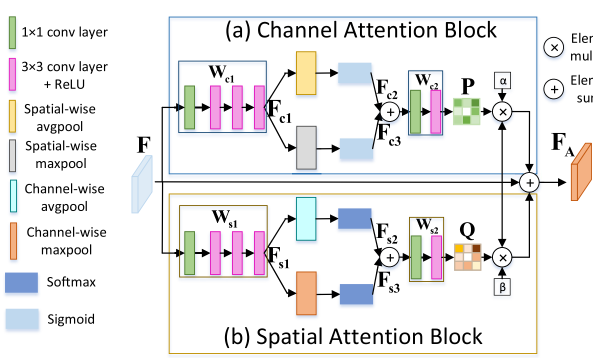

Fig. 4. Lesion feature attention module. It takes F as input, generates two attention maps P and Q, multiplied by F, and output the learned feature F_A.

recognizing the skin disease after segmenting and cropping out the central part of the image first [8].

However, these hand-crafted features are usually not robust, which struggle to extract deeper semantic information and suffer from intensive computation due to high dimensions.

TABLE III
RESULTS ON ISIC 2018 DATASET. (A) SHOWS THE RESULTS OF OUR METHOD AND OTHER SINGLE BACKBONE METHODS. (B) SHOWS THE RESULTS OF OUR METHOD AND OTHER METHODS ON LIVE LEADERBOARD. "BMCA" IS THE GOLD EVALUATION METRIC

(a)

Method	Category	$BMCA^*$	AUC	AP	ACC
Our CI-Net	-	**0.855**	0.970	**0.883**	**0.957**
L-CNN [23]	Skin lesion classification	0.844	**0.982**	0.874	0.944
CBAM [43]	Attention-based	0.840	0.968	0.874	0.950
Taxonomies [29]	Clinical-inspired for skin lesion classification	0.837	0.973	0.850	0.934
EfficientNet [57]	Natural image classification	0.837	0.971	0.863	0.923
MB-DCNN [22]	Skin lesion classification	0.832	0.945	0.857	0.939
DenseNet [58]	Natural image classification	0.825	0.980	0.860	0.925
SENet [42]	Attention-based	0.821	0.967	0.838	0.917
ResNet101 [59]	Natural image classification	0.812	0.941	0.834	0.929

(b)

Team	External Data	$BMCA^*$	AUC	Spe	Sen
#1	No	0.886	0.983	0.975	0.857
#2	No	0.875	0.953	**0.980**	0.819
Ours	No	0.873	0.985	**0.980**	**0.873**

TABLE IV
RESULTS ON ISIC 2019 DATASET. (A) SHOWS THE RESULTS OF OUR METHOD AND OTHER SINGLE BACKBONE METHODS. (B) SHOWS THE RESULTS OF OUR METHOD AND OTHER METHODS ON LIVE LEADERBOARD. "BMCA" IS THE GOLD EVALUATION METRIC

(a)

Method	Category	$BMCA^*$	AUC	AP	ACC
Our CI-Net	-	**0.657**	**0.882**	**0.541**	**0.849**
MB-DCNN [22]	Skin lesion classification	0.643	0.869	0.521	0.814
EfficientNet [57]	Natural image classification	0.639	0.871	0.531	0.835
SENet [42]	Attention-based	0.637	0.862	0.519	0.819
L-CNN [23]	Skin lesion classification	0.632	0.863	0.534	0.813
Taxonomies [29]	Clinical-inspired for skin lesion classification	0.632	0.867	0.512	0.840
DenseNet [58]	Natural image classification	0.629	0.858	0.519	0.802
CBAM [43]	Attention-based	0.624	0.861	0.525	0.821
ResNet101 [59]	Natural image classification	0.619	0.846	0.516	0.796

(b)

Team	External Data	$BMCA^*$	AUC	Spe	Sen
#1	No	0.645	0.868	0.951	0.712
Ours	No	0.640	**0.906**	**0.974**	0.538

Also, the discriminative ability is usually poor. Thus, it is no longer suitable and competitive against CNN-based method.

B. CNN-Based Methods for Skin Lesion Classification

On the other side, along with the development of deep learning, many CNN-based approaches have led to an overwhelming presence of the skin lesion classification problem.

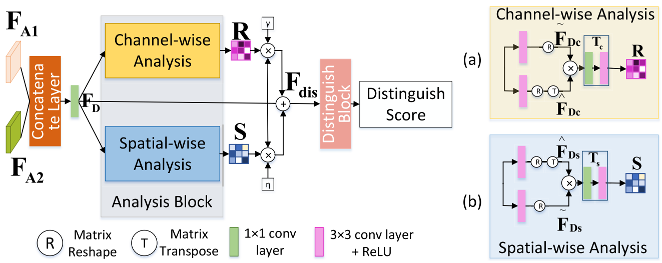

Fig. 5. Distinguish module. It takes F_{A1} and F_{A2} as input and output a distinguish score, which is the probability of two images belong to the same class.

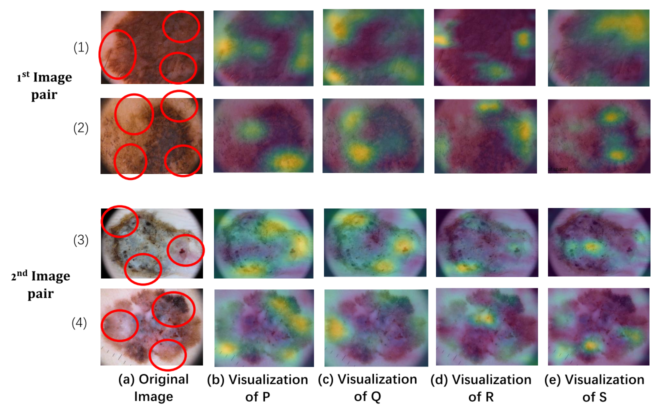

Fig. 6. Visualization results for melanoma. (a) is the original image, (b)-(e) are the visualization of attention maps P, Q, R and S respectively. The red circles denote regions matching the characteristics of melanoma, which is introduced in Section IV E. In each visualization result, regions with lighter colors denote large values in the attention map. The four attention maps successfully highlight the important regions, which are significant for the diagnosis.

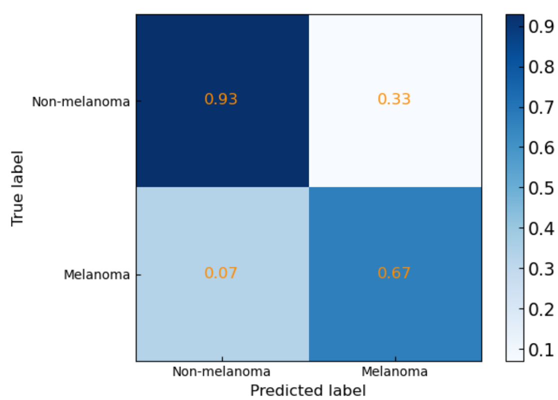

Fig. 7. Confusion matrix for ISIC 2016 dataset.

TABLE V

RESULTS OF OUR METHOD AND OTHER SINGLE BACKBONE METHODS ON ISIC 2020 DATASET. "AUC" IS THE GOLD EVALUATION METRIC

Method	Category	AUC^*	ACC	SE	SP
Our CI-Net	-	**0.923**	**0.960**	**0.322**	0.918
EfficientNet [57]	Natural image classification	0.910	0.952	0.201	**0.982**
Taxonomies [29]	Clinical-inspired for skin lesion classification	0.908	0.951	0.288	0.934
SENet [42]	Attention-based	0.908	0.948	0.139	0.901
L-CNN [23]	Skin lesion classification	0.903	0.950	0.279	0.917
MB-DCNN [22]	Skin lesion classification	0.900	0.943	0.176	0.943
CBAM [43]	Attention-based	0.894	0.937	0.150	0.923
DenseNet [58]	Natural image classification	0.884	0.934	0.317	0.901
ResNet101 [59]	Natural image classification	0.880	0.941	0.252	0.899

References [35], [36], and [37] provide specific reviews for these methods.

Specifically, SDL [15] introduces a dual network and a synergic loss to enable the two networks to facilitate each other. A new computer-aided diagnosis system is proposed in [38], including data augmentation, model fine-tuning, transfer learning and feature selection. MRCN [24] mines the relationship both the region level and image level relationship of dermoscopy images as for more comprehensive learning. A two-stream deep neural network information fusion framework is introduced in [39], which includes a fusion-based contrast enhancement module and a feature selection framework module for more discriminative feature generation.

Aside from dermoscopy images, clinical images, which are captured by the digital camera, are also widely used for clinical analysis. Thus, authors in [14] leverage deep residual network and bilinear pooling technique to represent global and local information of data from the two modalities, then aggregate the information for a more comprehensive classification. To avoid the overfitting problem since medical datasets usually suffer from insufficient data, a lightweight CNN is introduced in [23] based on the fine-grained classification principle. However, the generalization ability of these methods is not very good when the dataset is larger. To tackle this, some approaches design a deeper network to enhance the

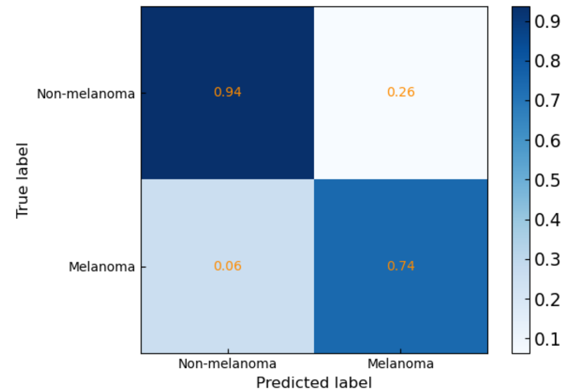

Fig. 8. Confusion matrix for ISIC 2017 dataset sub-task1.

TABLE VI

RESULTS ON PH2 DATASET. OUR METHOD PERFORMS BEST

Method	ACC	SE	SP	AUC
Our CI-Net	**0.950**	0.869	**0.960**	**0.983**
MB-DCNN [22]	0.940	0.95	0.938	0.977
CICS [60]	-	**1**	0.882	-
MFLF [61]	-	0.98	0.90	-
CCS [62]	-	0.925	0.763	0.843

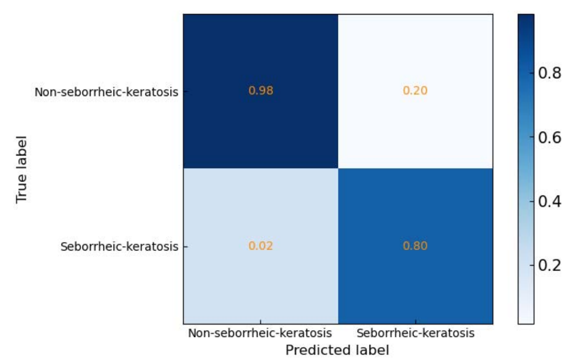

Fig. 9. Confusion matrix for ISIC 2017 dataset sub-task2.

TABLE VII

RESULTS ON ISIC 2017 DATASET. "AVERAGE AUC" IS THE GOLD EVALUATION METRIC. OUR CI-NET PERFORMS THE BEST

Methods	External data	Melanoma Classification					Seborrheic Keratosis					*Average*
		*AUC**	AP	ACC	SE	SP	*AUC**	AP	ACC	SE	SP	*AUC**
CI-Net	0	**0.950**	**0.868**	0.903	**0.802**	0.936	**0.989**	**0.920**	**0.950**	0.962	0.921	**0.970**
MRCN [24]	0	0.947	0.864	0.906	0.796	0.921	0.988	0.917	0.949	0.918	0.946	0.968
Teledermatology [52]	-	-	-	**0.926**	-	-	-	-	0.926	-	-	-
MB-DCNN [22]	1320	0.903	0.857	0.878	0.727	0.915	0.973	0.913	0.93	0.844	0.945	0.938
IRTL [55]	0	-	-	-	-	-	-	-	-	-	-	0.929
GP-CNN-DTEL [53]	0	0.891	-	0.850	0.376	0.965	0.960	-	0.935	0.722	0.933	0.926
SMDP [56]	0	-	-	-	-	-	-	-	-	-	-	0.923
DermaKNet [28]	900	0.873	-	-	-	-	0.962	-	-	-	-	0.917
ARL-CNN [21]	1320	0.875	-	0.850	0.658	0.896	0.958	-	0.868	0.878	0.867	0.917
SSAC [54]	1320	0.873	-	0.835	0.556	0.903	0.959	-	0.912	0.889	0.916	0.916
SDL [16]	1320	0.868	0.689	0.872	-	-	0.955	0.818	0.917	-	-	0.912
RENI [63]	1444	0.868	0.710	0.828	0.735	0.851	0.953	0.786	0.803	**0.978**	0.773	0.911
gpm-LSSSD [28]	900	0.856	0.747	0.823	0.103	**0.998**	0.963	0.839	0.875	0.178	**0.998**	0.910
Alea-Jacta-Est [64]	7544	0.874	0.715	0.872	0.547	0.950	0.943	0.790	0.895	0.356	0.990	0.908
EResNet [65]	1600	0.870	0.732	0.858	0.427	0.963	0.921	0.770	0.918	0.589	0.976	0.896
DLSL [66]	1341	0.836	0.665	0.845	0.350	0.965	0.935	0.808	0.913	0.556	0.976	0.886
Taxonomies [29]	0	0.800	-	-	-	-	0.935	-	-	-	-	0.868

TABLE VIII

STANDARD DEVIATION AND T-TEST AGAINST OTHER METHODS. THE IMPROVEMENT IS STATISTICALLY SIGNIFICANT

Dataset	Method (Ours vs)	t-value	p-value	Significant at p <0.05?	Standard deviation
ISIC2016	MRCN (rank2) L-CNN (rank3)	2.216(ACC) 2.869(ACC)	0.0273(ACC) 0.0044(ACC)	Yes Yes	0.0067
ISIC2017 sub-task1	MRCN (rank2) (rank3)	2.409(AP) 2.842(ACC)	0.0163(AP) 0.0046(ACC)	Yes Yes	0.0096
ISIC2017 sub-task2	MRCN (rank2) (rank3)	2.262(AP) 2.536(ACC)	0.0241(AP) 0.0115(ACC)	Yes Yes	0.0075
ISIC2018	CBAM (rank2) L-CNN (rank3)	2.493(ACC) 3.260(ACC)	0.0128(ACC) 0.0011(ACC)	Yes Yes	0.0024
ISIC2019	EfficientNet (rank2) MB-DCNN (rank3)	3.643(ACC) 8.971(ACC)	0.0003(ACC) p<0.0001(ACC)	Yes Yes	0.0054
ISIC2020	EfficientNet (rank2) Taxonomies (rank3)	4.293(ACC) 4.828(ACC)	p<0.0001(ACC) p<0.0001(ACC)	Yes Yes	0.0019
PH2	MB-DCNN (rank2) CICS (rank3)	3.313(ACC) 5.3380(ACC)	0.0011(ACC) p<0.0001(ACC)	Yes Yes	0.0086

TABLE IX

ANALYSIS OF COMPUTATION COMPLEXITY OF OUR CI-NET AND FOUR STATE-OF-THE-ART METHODS

Method	Type	Parameters	FLOPs
CI-Net (A) (Ours)	multi-stage	37.6M	20.5G
CI-Net (F) (Ours)		22.8M	14.8G
CI-Net (D) (Ours)		23.2M	14.9G
CI-Net (A+F) (Ours)		38.0M	20.8G
CI-Net (A+D) (Ours)		38.4M	20.9G
CI-Net (full) (Ours)		38.8M	21.2G
MRCN [24]	multi-stage	**73.8M**	36.8G
L-CNN [23]	single-stage	42M	23.2G
MB-DCNN [22]	multi-stage	54.9M	**62.8G**
GP-CNN-DTEL [53]	single-stage	37.2M	31.8G

TABLE X

ABLATION STUDY ON THE BACKBONE

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
ResNet18	**0.696**	**0.870**	**0.927**
ResNet34	0.692	0.841	0.914
ResNet50	0.689	**0.870**	0.921

representation capability. In DCNN-FV [19], Fisher Vector (FV) is adopted to take advantage of multiple visual representations for a more comprehensive analysis. A support vector machine with a Chi-squared kernel is also used to replace the fully connected layer for the classification. Nonetheless, those models are hard to train and require a large number of training strategies due to a larger number of parameters, which brings extra difficulties. Thus, some design multi-task models or extract multi-level features. To enable the network to focus

TABLE XI

ABLATION STUDY ON THE INPUT SIZE. (A) SHOWS THE RESULTS ON THE RESNET18 BACKBONE. (B) COMPARES OUR METHOD WITH STATE-OF-THE-ART METHODS

(a)

Method	Input size	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
ResNet18	224 × 224	0.667	0.848	0.909
ResNet18	336 × 336	0.676	0.862	0.910
ResNet18	448 × 448	**0.696**	**0.870**	**0.927**

(b)

Method	Input size	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)	PH2 (ACC)
CI-Net	224 × 224	**0.800**	**0.941**	**0.981**	**0.945**
MB-DCNN [22]	224 × 224	-	0.903	0.973	0.940
GP-CNN-DTEL [53]	224 × 224	0.728	0.891	0.960	-
CI-Net	336 × 336	0.806	0.946	0.983	0.945
CI-Net	448 × 448	**0.815**	**0.950**	**0.989**	**0.950**
MRCN [24]	448 × 448	0.807	0.947	0.988	-

on the region of interest, a two-branch network is proposed in [25], including a classification branch and a segmentation branch. The segmentation map is used to guide the classification network for better feature extraction. For simultaneous skin lesion segmentation and classification, a mutual deep convolutional neural network architecture is introduced in MB-DCNN [22]. An automated multi-task method is proposed in [40], which tackles skin lesion segmentation and classification simultaneously. It includes four steps: image enhancing, saliency estimation, feature extraction, and improved moth flame optimization for feature selection. Besides, to better utilize the data collected from dedicated hardware, a multi-modal information fusion framework is introduced in [41] to tackle both the segmentation and classification tasks. The segmentation result will be used for feature fusion to utilize the complementary strengths of two CNNs, which could also facilitate the overall result.

Although promising results have been achieved by these methods, they suffer from one major problem: did not involve medical expertise to enhance the interpretability. Certain protocols are required for the dermatologists to achieve this task, thus incorporating them is necessary, which is neglected by these methods. Due to this, the interpretability and usability are limited. Thus, they still have a large room to be improved.

C. Clinical Inspired Methods for Skin Lesion Classification

There are also a few works inspired by clinical expert knowledge [27], [28], [29], [30], [31]. Seven-point checklist [27] predicts the entire seven-point melanoma checklist,

TABLE XII
ABLATION STUDY FOR SHARING WEIGHTS

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
Without sharing weights	**0.813**	**0.950**	**0.989**
Sharing weights	0.803	0.937	0.980

TABLE XIII
ABLATION STUDY FOR THE NUMBER OF INPUT IMAGES N AND THE GPU MEMORY COST OF MODELS WITH DIFFERENT VALUES OF N

Method	ISIC2016 (AP)	GPU Memory Cost (GB)	ISIC2017 (AUC)	ISIC2016 (AUC)
CI-Net (N = 1)	5.7	0.751	0.881	0.934
CI-Net (N = 2)	9.3	**0.813**	**0.950**	**0.989**
CI-Net (N = 3)	14.4	0.780	0.938	0.984
CI-Net (N = 4)	16.6	0.766	0.933	0.976
CI-Net (N = 5)	19.3	0.762	0.927	0.975

TABLE XIV
ABLATION STUDY ON THE INFLUENCE OF THE TWO ATTENTION BLOCKS IN F

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
$F\_noCAnoSA$	0.713	0.884	0.940
F_CA	0.719	0.893	0.946
F_SA	0.728	0.887	0.945
F	**0.736**	**0.901**	**0.951**

TABLE XV
ABLATION STUDY ON THE INFLUENCE OF THE TWO ANALYSIS PARTS IN **D**

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
$D\_noCAnoSA$	0.709	0.894	0.933
D_CA	0.725	0.898	0.941
D_SA	0.715	0.900	0.939
D	**0.730**	**0.904**	**0.945**

TABLE XVI
ABLATION STUDY ON THE INFLUENCE OF THE CONV LAYERS FOR THE TWO ANALYSIS PARTS IN **D**

	Method	Whether adopting conv layer in channel-wise analysis	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
(a)	D	No	0.718	0.900	0.941
	D	Yes	**0.730**	**0.904**	**0.945**
	Method	Whether adopting conv layer in spatial-wise analysis	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
	D	No	0.725	0.899	0.942
	D	Yes	**0.730**	**0.904**	**0.945**
(b)	Method	Kernel size of conv layer in channel-wise analysis	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
	D	1 × 1	0.724	0.895	9.937
	D	3 × 3	**0.730**	**0.904**	**0.945**
	D	5 × 5	0.725	0.900	0.939
	D	7 × 7	0.723	0.899	0.940
	Method	Kernel size of conv layer in spatial-wise analysis	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
	D	1 × 1	0.727	0.901	0.942
	D	3 × 3	**0.730**	**0.904**	**0.945**
	D	5 × 5	0.727	0.899	0.941
	D	7 × 7	0.726	0.900	0.942

TABLE XVII
ABLATION STUDY ON THE INFLUENCE OF THE BINARY CROSS-ENTROPY LOSS IN **D**

Method	The loss for comparing the image pair	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
Ours	Baseline	0.696	0.870	0.927
Ours	Contrastive loss	0.721	0.896	0.933
Ours	Binary cross-entropy loss	**0.730**	**0.904**	**0.945**

which is widely used by doctors for clinical melanoma diagnosis. However, most of the public datasets do not have the annotation of these criteria since the annotation suffers from high costs. Thus, it poses a limitation for implementing this method on other skin diseases and other datasets automatically. Besides, this method ignores other medical expert knowledge for doctors' learning and diagnosis processes, such as segmenting the lesion area first and utilizing multiple inputs, etc. DermaKNet [28] argues to segment the lesion area first since it is commonly adopted by doctors. Besides, it also proposes polar pooling and asymmetry blocks, to mimic how dermatologists analyze skin lesions. Nevertheless, it does not utilize multiple images to explore the semantic correlations between different images, considering that multiple inputs could provide more information and facilitate the model by mutually learning from them [15], [16], [23]. Taxonomies [29] introduces a hierarchical structure, in order to mimic the hierarchical decision for doctors, as well as an attention module, to mimic the localized analysis for dermatologists. But, it just directly extracts the feature without segmenting the center lesion area first. Also, it only utilizes one image as input, ignoring to strengthen the network by capturing the information from multiple inputs, which is widely adopted by dermatology doctors. Our conference work CIN [32] mimics doctors' diagnostic and learning process, including three steps and one strategy respectively. However, the architecture is simple, which prevents it from deeply incorporating doctors' knowledge and learning better representations.

Existing clinical methods do incorporate some expert knowledge, they have not fully taken advantage of doctors' entire diagnosis process and learning strategy. Thus, there is still quite a room to be improved by including more clinical relevance and doctors' common knowledge in their learning and diagnosis process, and design a more interpretable method.

D. Attention Mechanism

Many works have been dedicated to introducing attention mechanisms into the network. For vision tasks based on natural images, to exploit relationships between channel-wise features, SENet [42] uses channel-wise multiplication between the attention weights. CBAM [43] proposes to utilize channel-wise and spatial-wise attention to boost the representation learned by the network. For skin lesion classification, ARL-CNN [21] integrates the attention mechanism with the residual network, including spatial and channel attention, to focus more on the valuable lesion area.

Different from previous works, which did not utilize the medical knowledge to provide better clinical interpretability, our method is motivated by an original perspective, that is, to mimic the subjective learning and diagnosis process of doctors.

TABLE XVIII
ABLATION STUDY ON THE INFLUENCE OF DIFFERENT HARD DATA MINING STRATEGY FOR IMAGE PAIR IN **D**

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
SS-LR [67]	0.719	0.889	0.934
BDR [68]	0.716	0.894	0.931
Ours	**0.730**	**0.904**	**0.945**

TABLE XIX
ABLATION STUDY ON THE INFLUENCE OF THE LOSS WEIGHT λ. THE RESULTS SHOW THAT SETTING λ TO 0.1 PERFORMS THE BEST

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
λ = 0.01	0.715	0.902	0.934
λ = 0.1	**0.730**	**0.904**	**0.945**
λ = 1	0.687	0.898	0.939
λ = 10	0.679	0.889	0.940

TABLE XX
ABLATION STUDY ON THE IMPACT OF EACH MODULE. THE RESULTS SHOW THE IMPACT OF EACH MODULE AND THEIR COMBINATION, WHICH ALL CONTRIBUTE TO THE IMPROVEMENT OF THE PERFORMANCE

Method	ISIC2016 (AP)	ISIC2017 sub-task1 (AUC)	ISIC2017 sub-task2 (AUC)
Baseline	0.696	0.870	0.927
CI-Net (A)	0.728	0.898	0.938
CI-Net (F)	0.736	0.901	0.951
CI-Net (D)	0.730	0.904	0.945
CI-Net (A+F)	0.770	0.929	0.969
CI-Net (A+D)	0.768	0.923	0.967
CI-Net (Full)	**0.813**	**0.950**	**0.989**

III. PROPOSED METHOD

As shown in Fig. 2, CI-Net contains two branches: the upper branch, and the lower branch, as well as four modules. The lesion area attention module is designed to simulate the **zoom step** of doctors' diagnostic process. It aims to segment the central lesion area and remove the irrelevant normal skin areas. The feature extraction module mimics the **observe step** of doctors' diagnostic process. It extracts the feature of an image. The lesion feature attention module simulates the **compare step** of dermatologists. Its goal is to mine the correlation of different local-attended sub-regions in the feature and locate the important ones, as to facilitate the classification. The distinguish module is motivated by doctors' **distinguish strategy** for their learning process. It aims to contrast images for achieving better representation learning and distinguish ability of the network. The following paper delves into the details.

A. Lesion Area Attention Module

1) Overview: This module simulates the **zoom step** of doctors' diagnostic process. It takes an image as input and crops out the central lesion area. So basically it conducts lesion segmentation on the original image. Note that the ground truth segmentation map is available to pre-train this module, which is illustrated specifically in Section IV B.

2) Specific Details: The experiment results in our conference paper [32] prove the effectiveness of this module. In this work, we adopt [44] to generate the lesion segmentation result. Then the smallest rectangle which includes the lesion area is used to crop out the image, which will be the output of this module. Meanwhile, the ground truth segmentation map and the generated lesion segmentation result are both utilized to crop the original image. Then a conv blocks, which contains two 3 × 3 conv layers, will extract the feature of these two cropped images. After that, L_1 loss will supervise the two features for extra regularization of the segmentation process.

B. Feature Extraction Module

This module mimics the **observe step** of doctors. A CNN backbone is adopted to extract the feature of the cropped images. We choose ResNet18 as the backbone network, which is also verified the best in Section IV-D.1. Denote the input image as I, the output feature would be $F \in \mathbb{R}^{C\times H\times W}$.

C. Lesion Feature Attention Module

1) Overview: This module mimics the **compare step** of dermatologists. It is equipped by each branch, takes F as input, generates two attention maps P and Q, multiplied by F, and outputs the learned feature F_A for each branch, denoted as F_{A1} and F_{A2}. The module consists of two blocks, as illustrated in Fig. 4. In the conference version [32], only spatial attention is conducted, i.e. the relationship between feature elements. In this paper, the architecture is extended to both spatial-wise and channel-wise, in order to strengthen the comprehensiveness of region information required by the compare step. The channel attention block takes F, analyzes the inter-channel relationship and outputs an attention map P. It adopts spatial-wise average-pooling and spatial-wise max-pooling to integrate the semantic information for each channel, then the importance of each channel and their relationship is further learned and highlighted by feature summation and multi-layer perceptrons (MLP). The spatial attention block takes F, models the spatial relationship and outputs an attention map Q. It adopts channel-wise average and max pooling, as well as MLPs to emphasize the effect of each element in the feature.

2) Specific Details: The specific architecture of channel attention block is illustrated in Fig. 4 (a). $F \in \mathbb{R}^{C\times H\times W}$ will first go through an MLP, denoted as W_{c_1}, for further representation learning, and obtains $F_{c_1} \in \mathbb{R}^{C\times H\times W}$. Then, F_{c_1} will be fed into two branches. To squeeeze the spatial information, a spatial-wise average pooling layer and a spatial-wise max pooling layer will be applied, followed by a sigmoid layer respectively for normalization. The outputs of these two branches are $F_{c_2} \in \mathbb{R}^{C\times 1}$ and $F_{c_3} \in \mathbb{R}^{C\times 1}$:

$$F_{c_2} = \rho(A_p(F_{c_1})), \qquad F_{c_3} = \rho(M_p(F_{c_1})) \tag{1}$$

where ρ denotes the sigmoid layer, A_p denotes the average pooling layer and M_p denotes the max pooling layer. After that, the summation of F_{c_2} and F_{c_3} will be fed into an MLP, denoted as W_{c_2}, to obtain the spatial attention map $P \in \mathbb{R}^{C\times 1}$, illustrated in Eqn. (2):

$$P = W_{c_2}(F_{c_2} + F_{c_3}) \tag{2}$$

Hence, important channels of the original feature F could be highlighted, resulting in higher values for the corresponding elements of P.

For the spatial attention block, the architecture is similar to the channel attention block, which is shown in Fig. 4 (b). F will first go through an MLP, denoted as W_{s_1}, and obtains $F_{s_1} \in \mathbb{R}^{C\times H\times W}$. Then, the channel-wise information will be aggregated by conducting channel-wise average pooling and a channel-wise max pooling respectively on F, followed by a softmax layer, to generate $F_{s_2} \in \mathbb{R}^{HW\times 1}$ and $F_{s_3} \in \mathbb{R}^{HW\times 1}$. After that, the summation of F_{s_2} and F_{s_3}, will be fed into an MLP, denoted as W_{s_2}, to generate the spatial attention map $Q \in \mathbb{R}^{HW\times 1}$, illustrated in Eqn. (3):

$$Q = W_{s_2}(F_{s_2} + F_{s_3}) \tag{3}$$

Therefore, important positions in the original feature F will be emphasized, resulting in higher values for the corresponding elements of Q.

After generating P and Q, element-wise multiplication was applied between P and F, and multiplied by a learnable weighting factor α. Similarly with Q. The equation is:

$$F_A = F + \alpha \cdot (P \cdot F) + \beta \cdot (Q \cdot F) \tag{4}$$

where "·" is element-wise multiplication.

Meanwhile, F_A will go through two fully connected layers, with a ReLU layer between them, and generate the lesion classification result. It is supervised by normal cross-entropy loss, which is denoted as L_c.

D. Distinguish Module

1) Overview: This module simulates **distinguish strategy**. In the conference version [32], this module only contains several learnable layers. In this paper, the architecture is improved by adding channel-wise analysis block and spatial-wise analysis block, to enhance the representation learning, which further improves the distinguish ability and facilitates the discrimination between categories. As illustrated in Fig. 5, it takes F_{A1} and F_{A2} as input and outputs a distinguish score, which is the probability of two images belonging to the same class. Three common sub-steps would be conducted by dermatologists to distinguish the two images: summarization of the features, analysis of further information, and conclusion drawing. On this basis, three sub-components are introduced in this module: a concatenate layer, an analysis block and a distinguish block. The concatenate layer concatenates the two input features for further learning. The analysis block aims at analyzing the characteristic between features of two images, thus it exploits the interdependencies between every two channels and spatial elements. The distinguish block consists of several learnable functions, to generate the distinguish score.

2) Specific Details: Firstly, F_{A1} and F_{A2} will be fed to the concatenated layer, and the concatenated feature will go through a 1×1 conv layer to further learn the semantic information and generate $F_D \in \mathbb{R}^{C\times H\times W}$, which will be the input of the following analysis block.

The analysis block contains two parts, the channel-wise analysis part and the spatial-wise analysis part. The channel-wise analysis part aims at exploiting the interdependencies between each channel of F_D, which is illustrated as Fig. 5 (a). F_D, the input, will go through a 3×3 conv layer and generates $\widetilde{F}_{Dc} \in \mathbb{R}^{C\times M}$, where M is equal to $H \times W$. Meanwhile, it goes through another 3×3 conv layer and obtains $\hat{F}_{Dc} \in \mathbb{R}^{M\times C}$. After that, to model the relation between every two channels, a matrix multiplication between $\widetilde{F}_{Dc}$ and $\hat{F}_{Dc}$ is applied, generating a matrix of size $C \times C$. The result is fed into an MLP, denoted as T_c, and obtains $R \in \mathbb{R}^{C\times C}$.

$$R = T_c(\widetilde{F}_{Dc} * \hat{F}_{Dc}) \tag{5}$$

where "*" means matrix multiplication. $R(i, j)$ represents the correlation between i^{th} channel and j^{th} channel in feature F_D.

The spatial-wise analysis part aims to model a wider range of contextual information between elements in F_D and enhance the representation capability. It has similar architecture to the channel-wise analysis part, as shown in Fig. 5 (b). F_D will pass through one 3×3 conv layer and generate $\hat{F}_{Ds} \in \mathbb{R}^{M\times C}$ and $\widetilde{F}_{Ds} \in \mathbb{R}^{C\times M}$ respectively. Then, to encode the correlation between channels, a matrix multiplication between $\hat{F}_{Ds}$ and $\widetilde{F}_{Ds}$ is applied. The result is fed into an MLP, denoted as T_s, obtaining $S \in \mathbb{R}^{M\times M}$:

$$S = T_s(\widetilde{F}_{Ds} * \hat{F}_{Ds}) \tag{6}$$

where "*" means matrix multiplication. $S(i, j)$ represents the correlation between i^{th} element and j^{th} element in F_D.

R will conduct a matrix multiplication with F_D, then the result is multiplied by a learnable weighting factor γ. Similarly with S and F_D. The output is F_{dis}, illustrated as:

$$F_{dis} = \gamma \cdot (R * F_D) + \eta \cdot (F_D * S) + F \tag{7}$$

where "*" is matrix multiplication.

Finally, F_{dis} goes through the distinguish block and conducts a deeper analysis for the two images. The output is a distinguish score, i.e., the probability of two images belonging to the same class. The distinguish score is supervised by the distinguish loss, which is a binary cross-entropy loss:

$$L_d = -\frac{1}{N_1}\sum_{i=1}^{N_1}(y_d^i log(x_d^i) + (1 - y_d^i) log(1 - x_d^i)) \tag{8}$$

where N_1 denotes the number of image pairs, y_d^i is the distinguish label of the i_{th} image pair, and x_d^i is the distinguish score, predicted by the module.

The final loss of the whole network is:

$$L = L_c + \lambda L_d \tag{9}$$

λ is a tunable parameter which balances the two losses.

IV. Experiments

A. Datasets

Six benchmark datasets are chosen for the experiments: the ISIC 2016 challenge dataset [45], which contains 1279 images from 2 classes; the ISIC 2017 challenge dataset [46], which contains 2750 images from 3 classes; the ISIC 2018 challenge dataset [47], [48], which contains 11720 images from 7 classes; the ISIC 2019 challenge dataset [46], [47], [48], which contains 33569 images from 8 classes; the ISIC 2020 challenge dataset [49], which contains 44108 images from 2 classes; the PH2 dataset [50], which contains 200 images from 2 classes. Table I illustrates the detail information of each dataset. Note that ISIC 2016 and ISIC 2017 are two ended challenges, ISIC 2018, ISIC 2019 and ISIC 2020 are ongoing challenges, in which the predictions of each image are needed to submit to the official platform [51] and obtain the performance results.

B. Implementation Details

ResNet18 is used as the backbone network for the feature extraction module. W_{c1} and W_{s1} are the combination of one 1×1 conv layer and three 3×3 conv layers. W_{c2} and W_{s2} consists of two 3×3 conv layers. T_c and T_s are the combination of a 1×1 conv layer and a 3×3 conv layer. The distinguish block of the distinguish module in Section III-D contains one 1×1 conv layer, two 3×3 conv layers and two fully connected layers, with ReLU and BN after each conv layer. The strategy of the training phase and the testing phase keep the same with our conference paper CIN [32], which has been verified effective. Specifically, for the training phase, we use the segmentation map for ISIC2016 to pre-train the lesion area attention module, which follows the setting of our conference work CIN [32] and other approaches such as MRCN [24] and MB-DCNN [22]. After that, two images are fed to lesion area attention module, which generates the cropped images, as the input of feature extraction module. The other modules are trained by Eq. 9. At the testing phase, a single image will be the input of the two branches. The average probability of the two branches is the final result. All the hyper-parameters are set by investigating the performance on the validation set. Since the ISIC 2016 dataset doesn't have a validation set, we randomly select 20% from the training set as the validation set. For the three ongoing challenge datasets, ISIC 2018, ISIC 2019 and ISIC 2020, we compare our method from two aspects: single backbone methods and methods on live leaderboard, for more comprehensive analysis. Each image will be resized to 448×448 and kept the same for training and testing. The base learning rate is set to 0.001 for all the datasets and reduced by 1/10 every 40 epochs. SGD algorithm is chosen as the optimizer, with a batch size of 64 on 8 NVIDIA RTX 2080Ti GPUs. For data augmentation, we conduct flip, scale, rotate, shift, and shear transformation. Specifically, scaling is conducted between 0.9 and 1.3, rotation is conducted from -90 to 90 degrees, shear transformation is conducted from -10 to 10 degrees, and shifting is conducted with 0.1. The image pairs are randomly chosen. In Eqn. (9), λ is set to 0.1. The following evaluation metrics are adopted to evaluate the datasets: average precision (AP), accuracy (ACC), area under the ROC curve (AUC), sensitivity (SE) and specificity (SP).

C. Benchmark Study

1) Performance on Each Dataset:

a) ISIC 2016 challenge dataset: We compare our performance to MRCN [24], L-CNN [23], DCNN-FV [19], SDL [15], DermaKNet [28], Taxonomies [29], Teledermatology [52], GP-CNN-DTEL [53] and top 5 methods of the challenge in 2016. We evaluate the performance on AP, ACC and AUC metric as to follow the tradition. Note that the challenge is ranked only by **AP**, which is the gold metric. DermaKNet and Taxonomies are two clinical inspired methods, and they do not report the results on ISIC 2016 dataset, so we used the code released by the authors for the experiment on this dataset. The other methods report their results, so we directly compare with them. The results are shown in Table II. We yield an AP of **0.815**, ACC of 0.904 and AUC of 0.904, which are all the best compared to other methods.

b) ISIC 2017 challenge dataset: We compare our performance to MRCN [24], MB-DCNN [22], ARL-CNN [21], SSAC [54], SDL [16], DermaKNet [28], Taxonomies [29], Teledermatology [52], GP-CNN-DTEL [53], IRTL [55], SMDP [56] and the top 5 methods on ISIC 2017 dataset. This challenge dataset contains two sub-tasks and the **Average AUC** of them is the gold metric. We evaluate the performance on AUC, AP, ACC, SE and SP and compare with other methods. The results are shown in Table VII. Our CI-Net achieves the best AUC on each sub-task, i.e., **0.950** and **0.989** respectively. The average AUC of our approach is **0.970**, which outperforms the second place. Besides, we also achieve the best performance of AP on two sub-tasks, ACC in sub-task2 and SE in sub-task1. The consistent advantage against other methods demonstrates the advancement of our model.

c) ISIC 2018 challenge dataset: For ISIC 2018, we compare our methods from two aspects: (1) single backbone methods; (2) methods on live leaderboard. The reason is that nearly all the methods on the live leaderboard adopt multiple backbones for model ensemble, which are not quite comparable with our method. Therefore, we first compare with several published methods which adopt a single backbone and conduct five-fold cross-validation to keep the same experiment setting for a fair comparison. After that, we conduct model ensemble and compare with the methods on the live leaderboard.

The single backbone methods are chosen from four aspects:

(1) SOTA methods for natural image classification, including ResNet [59], DenseNet [58] and EfficientNet [57].

(2) SOTA methods for skin lesion classification, including L-CNN [23], which performs the best on ISIC 2016, and MB-DCNN [22], which performs the best on ISIC 2017.

(3) Clinical-inspired method for skin lesion classification, including Taxonomies [29].

(4) Attention-based methods, including SENet [42] and CBAM [43].

Since the ground truth for the test set is unavailable, we conduct five-fold cross-validation on the training set. For all the compared methods, we use the code released by the authors. The gold metric for this dataset is the average recall score, denoted as **balanced multi-class accuracy (BMCA)**. The results are shown in Table III (a). Our method outperforms all the compared methods, with a BMCA of 0.855, improved the second place of 1.1%, and AP of 0.883 and ACC of 0.957. Also, we achieve a comparable AUC of 0.970.

Besides, we compare our method with other approaches on the live leaderboard. The results are shown in Table III (b). Our CI-Net ranks third among all submissions on the live leaderboard without additional data.

d) ISIC 2019 challenge dataset: Similar to ISIC 2018 dataset, we compare our method from two aspects.

Firstly, we compare with single backbone methods, which keep the same with ISIC 2018. As shown in Table IV (a), our CI-Net outperforms other methods in all the metrics.

Besides, we compare our method with other methods on the live leaderboard. The results are shown in Table IV (b). Our CI-Net ranks second among all submissions on the live leaderboard without additional data.

e) ISIC 2020 challenge dataset: Similar to ISIC 2018 and ISIC 2019 datasets, we conduct five-fold cross-validation on the training set of ISIC 2020. As shown in Table V, our CI-Net outperforms other methods in all the metrics.

f) PH2 Dataset: To follow the tradition of [22], we train the CI-Net on ISIC 2017 dataset, and then test on the entire PH2 dataset. We compare our performance to MB-DCNN [22], CICS [60], MFLS [61] and CCS [62]. The results are shown in Table VI. Our CI-Net yields the highest ACC with 0.950, SP of 0.960 and AUC of 0.983, and a comparable SE of 0.857.

2) T-Test and Standard Deviation Calculation for Each Dataset:

Besides, we conduct t-tests and calculate the standard deviation on each dataset against methods ranked second and third. As shown in Table VIII, the results confirm that the improvement of our method against other methods is statistically significant.

3) Comparison of Computation Complexity: We investigate the computational complexity of our CI-Net and four state-of-the-art methods with original code by authors, including MRCN [24], L-CNN [23], MB-DCNN [22] and GP-CNN-DTEL [53]. Note that MRCN ranks second in ISIC 2017 and third in ISIC 2016, MB-DCNN ranks second in PH2, GP-CNN-DTEL ranks fourth in ISIC 2016, L-CNN ranks fifth in ISIC 2016. Specifically, the number of parameters and FLOPs of the five methods are summarized in Table IX. The result reveals that our method contains parameters of 38.8M and FLOPs of 21.2G, which are the smallest no matter compared to single-stage or multi-stage methods. The reason may be that we adopt lightweight backbone models, and also keep the proposed modules lightweight. This demonstrates the effectiveness of our method, it achieves the best performance (shown in Table II to Table VII in Section IV C) while maintaining lightweight.

D. Ablation Study

Overview. Ablation study consists of three main aspects: the basic configurations of the model; the investigation of the component of individual modules (F and D) and the effect of all the proposed modules. It is conducted on three datasets: ISIC 2016, ISIC 2017 sub-task1 and ISIC 2017 sub-task2.

The three main modules of CI-Net are denoted as follows: 1. Lesion Area Attention Module (A); 2. Lesion Feature Attention Module (F); 3.The distinguish Module (D). The feature extraction module is used by default.

1) Basic Configurations of the Model: We first investigate the best setting of the model, including backbone, input size and whether sharing weights of the two branches. The best setting will be maintained in all the following experiments.

i. Backbone. Considering the balance between the number of parameters and the representation ability of the networks, we choose ResNet 18, ResNet34 and ResNet 50 for ablation study. Due to the small size of datasets, we do not choose large networks such as DenseNet because it could lead to over-fitting problem. Notice that modules A, F and D are not adopted in this experiment.

The results are shown in Table X. On ISIC 2016 dataset, ResNet18 achieves an AP of 0.696, compared to 0.692 and 0.689 of ResNet34 and ResNet50 respectively. Similarly on the other two datasets, ResNet18 outperforms the other two backbones, with an AUC of 0.870 and 0.927 respectively. Regarding the above results, ResNet18 is chosen as the backbone.

ii. Input size. We conduct experiments to evaluate the impact of different input sizes. The image is resized to 224×224, 336×336 and 448×448 respectively since they are commonly used in previous methods. The experiments are conducted on two models: the backbone model (ResNet18); and the full CI-Net model.

Table XI (a) shows the results of experiments on backbone model. The results reveal that the model with 448×448 input size yields the best performance on the three datasets, with an AP of 0.696 on ISIC 2016, AUC of 0.870 on ISIC 2017 sub-task1, and AUC of 0.927 on ISIC 2017 sub-task2. The model with 336×336 input size obtains results of the second place. Regarding the above results, 448×448 is chosen as the input size for the later experiments.

In Table XI (b), we compare our CI-Net with other state-of-the-art methods. We adopt our method under different input sizes on four datasets, including ISIC 2016, ISIC 2017 sub-task1, ISIC 2017 sub-task2, and PH2, and compare with three state-of-the-art methods: MRCN [24], MB-DCNN [22] and GP-CNN-DTEL [53]. Note that MRCN ranks second in ISIC 2017 and third in ISIC 2016, MB-DCNN ranks second in PH2, GP-CNN-DTEL ranks fourth in ISIC 2016. For the input size of 224×224, we compare our method with MB-DCNN [22] and GP-CNN-DTEL [53]. The results show that our method improves by a large margin on all datasets. Besides, for the input size of 448×448, our method outperforms MRCN [24] on all datasets. These prove the robustness of our method under different settings of input size.

iii. The impact of whether the two branches of the network share weight. We investigate whether sharing weights for the upper branch and the lower branch in Fig. 3 could influence performance. The experiment is conducted on the full model, where all the modules are adopted.

As shown in Table XII, on ISIC 2016 dataset, the model with sharing weights achieves a better AP of 0.813, compared to 0.803 for without sharing weights. Similarly, on the other two datasets, the model with sharing weights obtains an AUC of 0.950 and 0.989, with an improvement of 1.3% and 0.9%.

Besides, in our model, ResNet18 is chosen as the backbone, which has about 11.21M parameters. On the contrary, most of the methods choose ResNet50 as the backbone, the number of parameters for which is about 23.64M, which is more than 2 times of our backbone, meaning that even if we implement two separate models without sharing weights, our model still has a smaller number of parameters.

The results prove that adopting two branches without sharing weights is better, which is also more lightweight than other state-of-the-art methods.

iv. the influence of the number of input images. We investigate the best setting for the number of input images. The experiment is conducted on the full model, where all the modules are adopted. The number of the input images is denoted as N, from 2 to 5.

As shown in Table XIII, when N is equal to 2, the model performs the best, with an AP of 0.813 on ISIC 2016, AUC of 0.950 on ISIC 2017 sub-task1, and AUC of 0.989 on ISIC 2017 sub-task2 respectively. On ISIC 2017 sub-task2, the performance between $N = 2$ and $N = 3$ is close, but still, $N = 2$ is slightly better.

Meanwhile, we calculate the GPU memory cost of models under 448×448 input size when N is set to different values. As shown in Table XIII, along with the increase of N, the memory cost increases gradually. When N is larger than 2, the model would not be able to perform on one GPU, which poses a greater demand for computing resources.

To sum up, in consideration of the trade-off between the performance and the computational complexity, we suggest setting N to 2, and 3 if there is more than one GPU.

2) The Investigation of the Individual Modules: We use the best setting in Section IV-D.1 for the following experiments.

i. The influence of the sub-components for F and D.

We investigate the impact of the sub-components for F and D. First, we only adopt F and its variations on the backbone model (ResNet18), to investigate its sub-components. The results are shown in Table XIV. "$F\_noCAnoSA$" means without adopting any attention block. "F_CA" means only using the channel attention block as this module and combined with the baseline. "F_SA" denotes only adopting the spatial attention block as this module. Note that for "$F\_noCAnoSA$", the input feature will simply go through an MLP, which has the same architecture as W_{c1}, and generates the output.

"$F\_noCAnoSA$" achieves AP of 0.713, AUC of 0.884 and AUC of 0.940 respectively on the three datasets. After adopting "CA" and "SA", the performance further improves to 0.719 and 0.728 respectively on ISIC 2016 dataset. Similarly with the other two datasets. Finally, "F" yields an AP of 0.736, an AUC of 0.901 and an AUC of 0.951 respectively on the three datasets. The results show that each block of F is effective and contributes to the improvement of performance.

Similarly, the investigation is applied for D on the backbone model (ResNet18). "$D\_noCAnoSA$" means without adopting any analysis part. "D_CA" denotes that only adopting channel-wise analysis in this module and combined with the baseline. "D_SA" means only conducting spatial-wise analysis in this module.

The results are shown in Table XV. "$D\_noCAnoSA$" obtains an AP of 0.709, AUC of 0.894 and AUC of 0.933 on the three datasets respectively. After adopting "CA" and "SA", the performance further improves to 0.725, 0.715 respectively on ISIC 2016 dataset. Similarly with the other two datasets. Finally, "D" obtains an AP of 0.730, an AUC of 0.904, and an AUC of 0.945 on the three datasets. The performance shows that each analysis benefits the network.

ii. The impact of the convolutional layers for D.

We conduct experiments to evaluate the impact of the convolutional layers used in channel-wise analysis part and spatial-wise part analysis for D. The experiments include two aspects: whether adopting them is necessary; the effect of different convolutional kernel sizes. In Table XVI (a), the results show that for both the two analysis parts, the convolutional layers could effectively learn the semantic information and facilitate the correlation modeling. In Table XVI (b), the results show that for both the two analysis parts 3×3 is the best setting for the convolutional layers.

iii. The impact of the binary cross-entropy for D.

We conduct experiments to evaluate the impact of two different losses for comparing the image pair in D. The first is the binary cross-entropy loss, which is the originally adopted loss. The second loss is the contrastive loss. Specifically, for an image pair, if they are from the same class, they will be denoted as "positive pair", otherwise they will be denoted as "negative pair". The results are shown in Table XVII. They indicate that both the two losses could facilitate the performance, but the cross-entropy loss performs better. The reason for cross-entropy loss performing better may be that for few image pairs, they are from the same class, but their representation is dissimilar, due to the intra-class variation of this task. In this case, the contrastive loss will force the representation to be similar, which may instead weaken the discrimination ability. On the other hand, no matter what the class label, the cross-entropy loss will force the network to distinguish whether the two images in a pair are from the same class, which won't be influenced by the above circumstance. But generally, both the two losses could improve the performance, compared to not adopting any.

iv. The impact of data mining strategy for image pair in D.

We conduct experiments to evaluate the impact of different hard data mining strategies for image pairs in D. We compare with two hard mining methods, SS-LR [67], which aims to obtain a compact yet representative background data points to improve the image classifier, and BDR [68], which introduces two loss functions to tackle the negative class distraction problem, to see the impact of their data mining strategy. The results are shown in Table XVIII. It indicates that the extra mining strategies in the two methods did not bring extra improvement to the final performance. This may be because for skin disease, the metric to measure the similarity of the representation of two features may not be quite in line with which in SS-LR and BDR, thus the strategies in these two methods are not suitable for our task, which hurt the performance. The result also reveals that for skin disease, it is worth studying the data mining strategy, as to select image pairs for more effective comparison and discrimination.

v. The weight of the distinguish loss. We investigate the performance of different values for λ respectively: 0.01, 0.1, 1 and 10. Only D is adopted on the backbone model (ResNet18) to conduct the experiment, since the distinguish loss is generated by D. The results are shown in Table XIX.

On ISIC 2016 dataset, when λ is equal to 0.1, the performance is the best, with an AP of 0.730 and an obvious improvement of 1.5%. Similarly for the other two datasets. When λ is set to 0.1, the model obtains an AUC of 0.904 and 0.945. The performance is close but slightly better than that of the other three values. Regarding the results, 0.1 is the best setting for λ among all the selected values.

3) The Effect of All the Proposed Modules: We investigate the influence of adding each module and combination of them on the backbone (ResNet18). The results are shown in Table XX. For simplicity, we remove the "CI-Net" in the following paper. For example, "CI-Net(A)" is replaced with "(A)".

On ISIC 2016 challenge dataset, we first evaluate the performance of adding three modules to the baseline, denoted as "(A)", "(F)" and "(D)" respectively. Compared to the baseline, the performance of AP value is improved by 1.1%, 1.6% and 2.1%, respectively. This proves the effectiveness of each module. Next, when two modules are adopted, "$(A+F)$" obtains an AP of 0.770, with a gain of 4.2% and 3.4% respectively compared with "(A)" and "(F)". For "$(A+D)$", the AP is improved by 4.0% and 3.8% respectively. Finally, when we adopt the full model, denoted as "$(Full)$", the model yields an AP of 0.813, which is 11.7% higher than that from the "$(Baseline)$". Similarly for the other two datasets, the results demonstrate the effectiveness of each module and the combination of them, and reveal the successfulness of our migration from the knowledge of doctors to CNN.

E. Clinical Protocol and Visualization Analysis

To provide better interpretability, we utilize the clinical protocol of skin disease to explain the visualization result, in order to qualitatively demonstrate the clinical effectiveness of our method.

Since all the six benchmark datasets contain "melanoma", we introduce the clinical protocol for diagnosing this disease. We first introduce the protocol, then utilize specific images to illustrate it. According to [69], which is very authoritative on the clinical application of dermoscopy, there are several important features for diagnosing melanoma: 1. asymmetry of structure; 2. uneven pigment distribution; 3. blue-white structures. The red circles in Fig. 6 (a) denote sub-regions that meet these three features. Let (1), (2), (3), (4) represent the four image samples respectively. Specifically, for the first feature, in column (a) of Fig. 6, regions circled by the two right circles in (1), and the two right circles in (2) show the feature. For the second feature, in column (a) of Fig. 6, regions circled by the left circle in (1), the two left circles in (2), the top left and the right circles in (3), the left and the bottom right circles in (4) show the feature. For the third feature, in column (a) of Fig. 6, regions circled by bottom left circle in (3) and the top right circle in (4) show the feature.

To verify whether our method meets the clinical protocol, we visualize the attention maps of P, Q, R and S, which are served for locating meaningful sub-regions. The results are shown in Fig. 6. (a) is the original images, (b)-(e) are the visualization of P, Q, R and S respectively. In each visualization result, regions with lighter colors denote large values in the attention map. It can be seen that the activation result of the four attention maps successfully highlights and pays attention to the important and diagnosis-related regions mentioned above. These results show that our method could learn useful clinical information for better representation encoding.

F. Confusion Matrix Result

The confusion matrices for ISIC 2016 and ISIC 2017 datasets are presented in Fig. 7, Fig. 8 and Fig. 9. They provide detailed information about correct and incorrect predictions. For ISIC 2016 dataset, we can see that the accuracy of melanoma classification is only 0.67. On the other hand, the accuracy of classifying non-melanoma is 0.93. This indicates that the discrimination against melanoma is more difficult. The reason may be the imbalance dataset, which means that the number of samples for melanoma is smaller than the other category. For ISIC 2017 dataset sub-task1, the accuracy of melanoma classification is 0.74. Meanwhile, the accuracy of classifying non-melanoma is 0.94. This also reveals that the classification of melanoma is more challenging. For ISIC 2017 dataset sub-task2, the accuracy of seborrheic keratosis is 0.80. On the other hand, the accuracy of classifying non-seborrheic-keratosis is 0.98. This indicates that the discrimination of seborrheic keratosis is better than melanoma.

G. Limitation of Our Work

The main limitation of our work is two-fold. Firstly, the random choice of image pairs may lead to ineffective image comparison. Specifically, if both images are very different, their representation will also be very different, then the comparison may not be useful and effective. Thus, for more effective pair selection and comparison, hard example mining strategies are needed. Secondly, the architecture of attention mechanisms in lesion feature attention module and distinguish module is adopted in a self-supervised way, which may ineffectively model the correlation for sub-regions and encode their representation in some cases. Thus, stronger supervision for more solid correlation learning is needed, which means that the analysis could go beyond just attention mechanisms.

V. Conclusion

In this work, to fully take advantage of the knowledge which doctors commonly conduct during their learning and the diagnosis process, we present a clinical-inspired network

(CI-Net) to simulate these processes. Specifically, a lesion area attention module, a feature extraction module and a lesion feature attention module is introduced to mimic the three steps of diagnosis process: the zoom step, the observe step and the compare step. Besides, to involve the distinguish strategy in doctors' learning process, we design a distinguish module.

Compared to other clinical-inspired methods, such as Seven-point checklist [27], DermaKNet [28], Taxonomies [29], CSLDR [30], DermESy [31], our method incorporates the knowledge of doctors' entire diagnosis process and learning strategy. To mimic the expertise step by step, we design four novel modules, which are equipped with better representation learning ability, compared to previous approaches. Besides, we provide clinical protocol, and utilize the visualization result to qualitatively show the clinical interpretability and usability of our method. The state-of-the-art experiment results on six benchmark datasets, both quantitatively and qualitatively, verify the successful migration from medical expertise to neural networks.

In future works, there are two potential directions for this task. Firstly, for the choice input pair, the research of hard example sampling strategy, in order to select image pairs for more effective comparison, is worth studying. Secondly, how to mine a deeper correlation with stronger supervision and clinical interpretability beyond utilizing current attention mechanisms within and between representations of images, e.g., considering more specific knowledge and information about the diseases, is also worth researching.

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